27. Watchdog Timer (WDT)

27.1 Overview

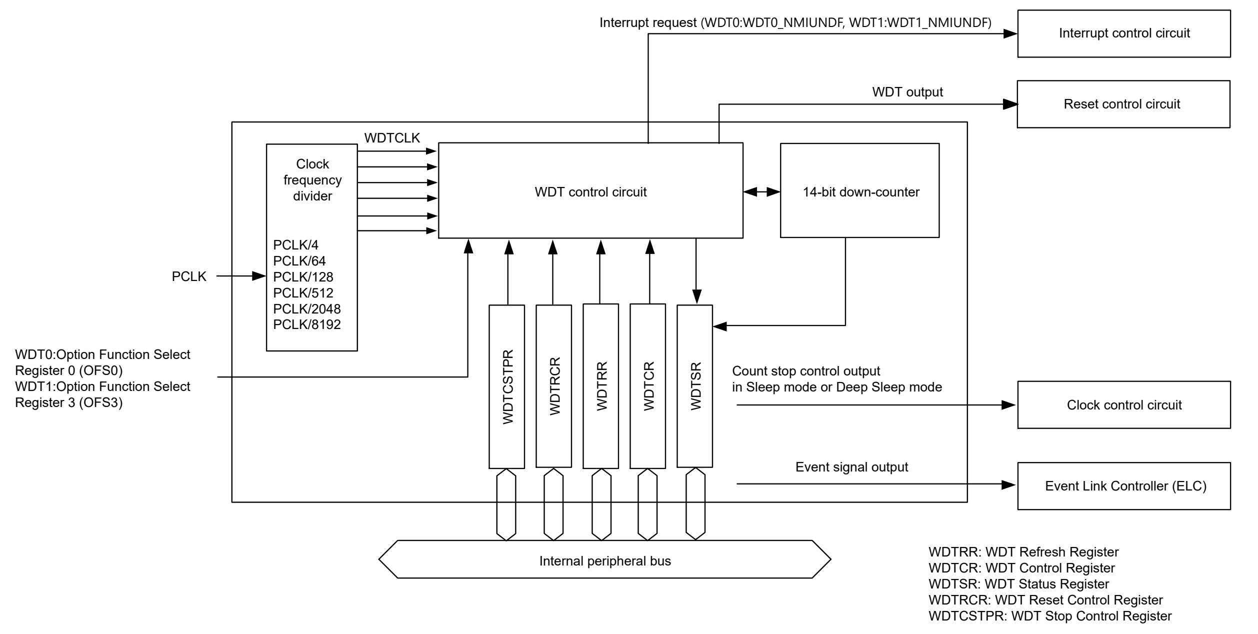
The Watchdog Timer (WDT) is a 14-bit down counter that can be used to reset the MCU when the counter underflows because the system has run out of control and is unable to refresh the WDT. In addition, the WDT can be used to generate a non-maskable interrupt or an underflow interrupt.

Table 27.1 lists the WDT specifications and Figure 27.1 shows a block diagram.

Table 27.1 WDT specifications

Parameter	Specifications
Number of units	2 (WDT0 for CPU0, WDT1 for CPU1)
Count source[*1]	Peripheral clock (PCLKB)
Clock division ratio	Division by 4, 64, 128, 512, 2048, or 8192
Counter operation	Counting down using a 14-bit down-counter
Condition for starting the counter	● Auto start mode: Counting automatically starts after a reset or after an underflow or refresh error occurs ● Register start mode: Counting is started with a refresh by writing to the WDTRR register ● Only secure developer can select Auto-start mode or Register-start mode
Conditions for stopping the counter	● Reset (the down-counter and other registers return to their initial values) ● A counter underflows or a refresh error is generated
Window function	Window start and end positions can be specified (refresh-permitted and refresh-prohibited periods)
Watchdog timer reset sources	● Down-counter underflows ● Refreshing outside the refresh-permitted period (refresh error)
Non-maskable interrupt/interrupt sources	● Down-counter underflows ● Refreshing outside the refresh-permitted period (refresh error)
Reading of the counter value	The down-counter value can be read by the WDTSR register
Event link function (output)	● Down-counter underflow event output ● Refresh error event output
Output signal (internal signal)	● Reset output ● Interrupt request output ● Sleep mode or Deep Sleep mode count stop control output
TrustZone Filter	Security and Privilege attribution can be set

Note 1. Satisfy the frequency of the peripheral module clock (PCLKB) ≥ 4 × (the frequency of the count clock source after division).



Note: PCLK = PCLKB

Figure 27.1 WDT block diagram

27.2 Register Descriptions

27.2.1 WDTRR : WDT Refresh Register

Base address: WDTn = 0x4020_2600 + 0x0100 × n (n = 0, 1)
WDTn_NS = 0x5020_2600 + 0x0100 × n (n = 0, 1)

Offset address: 0x00

Bit position:	7							0
Bit field:								
Value after reset:	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	n/a	The down-counter is refreshed by writing 0x00 and then writing 0xFF to this register.	R/W

Note: S-TYPE-3, P-TYPE-3

The WDTRR register refreshes the down-counter of the WDT.

The down-counter of the WDT is refreshed by writing 0x00 and then writing 0xFF to WDTRR register (refresh operation) within the refresh-permitted period.

After the down-counter is refreshed, it starts counting down from the value selected by setting the WDT Timeout Period Select bits (OFS0.WDT0TOPS[1:0] or OFS3.WDT1TOPS[1:0]) in the Option Function Select Register m in auto start mode. In register start mode, counting down starts from the value selected by setting the Timeout Period Select bits (WDTCR.TOPS[1:0]) in the WDT Control Register.

When 0x00 is written, the read value is 0x00. When a value other than 0x00 is written, the read value is 0xFF. For details of the refresh operation, see section 27.3.3. Refresh Operation.

27.2.2 WDTCR : WDT Control Register

Base address: WDTn = 0x4020_2600 + 0x0100 × n (n = 0, 1)
WDTn_NS = 0x5020_2600 + 0x0100 × n (n = 0, 1)

Offset address: 0x02

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RPSS[1:0]		—	—	RPES[1:0]		CKS[3:0]				—	—	TOPS[1:0]	
Value after reset:	0	0	1	1	0	0	1	1	1	1	1	1	0	0	1	1

Bit	Symbol	Function	R/W
1:0	TOPS[1:0]	Timeout Period Select 0 0: 1024 cycles (0x03FF) 0 1: 4096 cycles (0x0FFF) 1 0: 8192 cycles (0x1FFF) 1 1: 16384 cycles (0x3FFF)	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
7:4	CKS[3:0]	Clock Division Ratio Select 0 0 0 1: PCLKB/4 0 1 0 0: PCLKB/64 1 1 1 1: PCLKB/128 0 1 1 0: PCLKB/512 0 1 1 1: PCLKB/2048 1 0 0 0: PCLKB/8192 Others: Setting prohibited	R/W
9:8	RPES[1:0]	Window End Position Select 0 0: 75% 0 1: 50% 1 0: 25% 1 1: 0% (do not specify window end position).	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	RPSS[1:0]	Window Start Position Select 0 0: 25% 0 1: 50% 1 0: 75% 1 1: 100% (do not specify window start position).	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The WDTCR register is used to set the clock division ratio, and window start and end positions for refresh, and the timeout period until the down-counter underflows in register start mode.

Some constraints apply to writes to the WDTCR register. For details, see section 27.3.2. Controlling Writes to the WDTCR, WDTRCR, and WDTCSTPR Registers.

In auto start mode, the settings in the WDTCR register are disabled, and the settings in the Option Function Select Register m (OFSm) are enabled (m = 0, 3). The settings for the WDTCR register can also be made in the OFSm register (m = 0, 3). For details, see section 27.3.8. Association Between Option Function Select Register m (OFSm) and WDT Registers (m = 0, 3).

TOPS[1:0] bits (Timeout Period Select)

The TOPS[1:0] bits select the timeout period, the period until the down-counter underflows, from 1024, 4096, 8192, and 16384 cycles, taking the divided clock specified in the CKS[3:0] bits as 1 cycle. After the down-counter is refreshed, the combination of the CKS[3:0] and TOPS[1:0] bits determines the number of PCLKB cycles until the counter underflows.

Table 27.2 lists the relationship between the CKS[3:0] and TOPS[1:0] bit settings, the timeout period, and the number of PCLKB cycles.

Table 27.2 Timeout period settings

CKS[3:0] bits	TOPS[1:0] bits	Clock division ratio	Timeout period (number of cycles)	PCLKB clock cycles
0x1	00b	PCLKB/4	1024	4096
	01b		4096	16384
	10b		8192	32768
	11b		16384	65536
0x4	00b	PCLKB/64	1024	65536
	01b		4096	262144
	10b		8192	524288
	11b		16384	1048576
0xF	00b	PCLKB/128	1024	131072
	01b		4096	524288
	10b		8192	1048576
	11b		16384	2097152
0x6	00b	PCLKB/512	1024	524288
	01b		4096	2097152
	10b		8192	4194304
	11b		16384	8388608
0x7	00b	PCLKB/2048	1024	2097152
	01b		4096	8388608
	10b		8192	16777216
	11b		16384	33554432
0x8	00b	PCLKB/8192	1024	8388608
	01b		4096	33554432
	10b		8192	67108864
	11b		16384	134217728

CKS[3:0] bits (Clock Division Ratio Select)

The CKS[3:0] bits specify the division ratio of the clock used for the down-counter. The division ratio can be selected from the PCLKB divided by 4, 64, 128, 512, 2048, and 8192. Combined with the TOPS[1:0] bit setting, this allows the WDT to be configured to a count period between 4096 and 134217728 PCLKB clock cycles.

RPES[1:0] bits (Window End Position Select)

The RPES[1:0] bits specify the window end position that indicates the refresh-permitted period. 75%, 50%, 25%, or 0% of the timeout period can be selected for the window end position. Set the window end position to a value less than the value for the window start position (window start position > window end position). If the window start position is set to a value less than or equal to the window end position, the window start position setting is enabled, and the window end position is set to 0%.

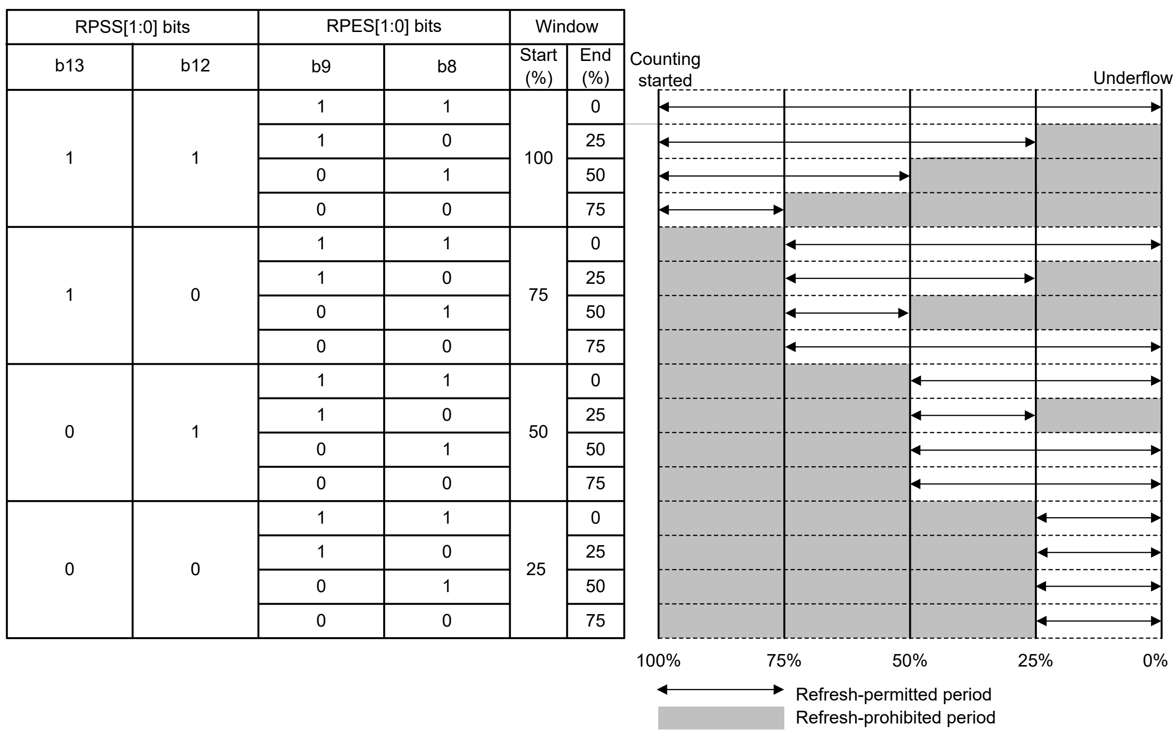
RPSS[1:0] bits (Window Start Position Select)

The RPSS[1:0] bits specify the window start position that indicates the refresh-permitted period. 100%, 75%, 50%, or 25% of the timeout period can be selected for the window start position. Set the window start position to a value greater than the value for the window end position. If the window start position is set to a value less than or equal to the window end position, the window start position setting is enabled, and the window end position is set to 0%.

Table 27.3 lists the counter values for the window start and end positions, and Figure 27.2 shows the refresh-permitted period set in the RPSS[1:0], RPES[1:0], and TOPS[1:0] bits.

Table 27.3 Relationship between the timeout period and window start and end counter values

TOPS[1:0]	Timeout period		Window start and end counter value			
	Cycles	Counter value	100%	75%	50%	25%
00b	1024	0x03FF	0x03FF	0x02FF	0x01FF	0x00FF
01b	4096	0x0FFF	0x0FFF	0x0BFF	0x07FF	0x03FF
10b	8192	0x1FFF	0x1FFF	0x17FF	0x0FFF	0x07FF
11b	16384	0x3FFF	0x3FFF	0x2FFF	0x1FFF	0x0FFF



Note: If window end position setting ≥ window start position setting, the window end position setting is set to 0%.

Figure 27.2 RPSS[1:0] and RPES[1:0] bits setting and refresh-permitted period

27.2.3 WDTSR : WDT Status Register

Base address: WDTn = 0x4020_2600 + 0x0100 × n (n = 0, 1)
WDTn_NS = 0x5020_2600 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	REFEF	UNDFF	CNTVAL[13:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	CNTVAL[13:0]	Down-Counter Value Value counted by the down-counter	R
14	UNDFF	Underflow Flag 0: No underflow occurred 1: Underflow occurred	R/W*1

Bit	Symbol	Function	R/W
15	REFEF	Refresh Error Flag 0: No refresh error occurred 1: Refresh error occurred	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. Only 0 can be written to clear the flag.

The WDTSR register indicates the counter value of the down-counter and the status of whether an underflow or refresh error occurred in the down-counter.

CNTVAL[13:0] bits (Down-Counter Value)

Read the CNTVAL[13:0] bits to confirm the value of the down-counter. The read value might differ from the actual count by 1.

UNDFF flag (Underflow Flag)

Read the UNDFF flag to confirm whether an underflow occurred in the counter. A value of 1 indicates that the down counter underflowed. Write 0 to the flag to set the value to 0. Writing 1 has no effect.

Clearing of the UNDFF flag takes (N + 1) PCLKB cycles. In addition, clearing of the flag is ignored for (N + 2) PCLKB cycles after an underflow. N is specified as follows:

Auto start mode:

- When OFSm.WDTnCKS[3:0] = 0x1, N = 4
- When OFSm.WDTnCKS[3:0] = 0x4, N = 64
- When OFSm.WDTnCKS[3:0] = 0xF, N = 128
- When OFSm.WDTnCKS[3:0] = 0x6, N = 512
- When OFSm.WDTnCKS[3:0] = 0x7, N = 2048
- When OFSm.WDTnCKS[3:0] = 0x8, N = 8192

Register start mode:

- When WDTCR.CKS[3:0] = 0x1, N = 4
- When WDTCR.CKS[3:0] = 0x4, N = 64
- When WDTCR.CKS[3:0] = 0xF, N = 128
- When WDTCR.CKS[3:0] = 0x6, N = 512
- When WDTCR.CKS[3:0] = 0x7, N = 2048
- When WDTCR.CKS[3:0] = 0x8, N = 8192

REFEF flag (Refresh Error Flag)

Read the REFEF flag to confirm whether a refresh error occurred, indicating that a refresh operation was performed during a prohibited period. A value of 1 indicates that a refresh error occurred. Write 0 to the flag to set the value to 0. Writing 1 has no effect.

Clearing of the REFEF flag takes (N + 1) PCLKB cycles. In addition, clearing of the flag is ignored for (N + 2) PCLKB cycles after a refresh error. N is specified as follows:

Auto start mode:

- When OFSm.WDTnCKS[3:0] = 0x1, N = 4
- When OFSm.WDTnCKS[3:0] = 0x4, N = 64
- When OFSm.WDTnCKS[3:0] = 0xF, N = 128
- When OFSm.WDTnCKS[3:0] = 0x6, N = 512
- When OFSm.WDTnCKS[3:0] = 0x7, N = 2048
- When OFSm.WDTnCKS[3:0] = 0x8, N = 8192

Register start mode:

- When OFSm.WDTnCKS[3:0] = 0x1, N = 4
- When OFSm.WDTnCKS[3:0] = 0x4, N = 64
- When OFSm.WDTnCKS[3:0] = 0xF, N = 128
- When OFSm.WDTnCKS[3:0] = 0x6, N = 512
- When OFSm.WDTnCKS[3:0] = 0x7, N = 2048
- When OFSm.WDTnCKS[3:0] = 0x8, N = 8192

27.2.4 WDTRCR : WDT Reset Control Register

Base address: WDTn = 0x4020_2600 + 0x0100 × n (n = 0, 1)
WDTn_NS = 0x5020_2600 + 0x0100 × n (n = 0, 1)

Offset address: 0x06

Bit position:	7	6	5	4	3	2	1	0
Bit field:	RSTIRQS	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	RSTIRQS	WDT Behavior Selection 0: Interrupt 1: Reset	R/W

Note: S-TYPE-3, P-TYPE-3

The WDTRCR register controls reset output by a WDT down-counter underflow or interrupt request output.

Some constraints apply to writes to the WDTRCR register. For details, see section 27.3.2. Controlling Writes to the WDTCR, WDTRCR, and WDTCSTPR Registers.

In auto start mode, the WDTRCR register settings are disabled, and the settings in the Option Function Select register m (OFSm) are enabled (m = 0, 3). The settings for the WDTRCR register can also be made for the OFSm register (m = 0, 3). For details, see section 27.3.8. Association Between Option Function Select Register m (OFSm) and WDT Registers (m = 0, 3).

27.2.5 WDTCSTPR : WDT Count Stop Control Register

Base address: WDTn = 0x4020_2600 + 0x0100 × n (n = 0, 1)
WDTn_NS = 0x5020_2600 + 0x0100 × n (n = 0, 1)

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SLCSTP	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	SLCSTP	Sleep-Mode Count Stop Control Register 0: Disable count stop 1: Stop count on transition to Sleep mode or Deep Sleep mode	R/W

Note: S-TYPE-3, P-TYPE-3

The WDTCSTPR register controls whether to stop the WDT counter in Sleep mode or Deep Sleep mode. Some constraints apply to writes to the WDTCSTPR register. For details, see section 27.3.2. Controlling Writes to the WDTCR, WDTRCR, and WDTCSTPR Registers.

In auto start mode, the WDTCSTPR register settings are disabled, and the settings in the Option Function Select register m (OFSm) are enabled (m = 0, 3). The settings for the WDTCSTPR register can also be made for the OFSm register (m = 0, 3). For details, see section 27.3.8. Association Between Option Function Select Register m (OFSm) and WDT Registers (m = 0, 3).

SLCSTP bit (Sleep-Mode Count Stop Control Register)

The SLCSTP bit of WDT0 selects whether to stop counting on CPU0 transition to Sleep mode or Deep Sleep mode.

The SLCSTP bit of WDT1 selects whether to stop counting on CPU1 transition to Sleep mode or Deep Sleep mode.

27.2.6 Option Function Select Register

For information on the OFS0 register and the OFS3 register, see section x, Option-Setting Memory.

27.3 Operation

27.3.1 Count Operation in each Start Mode

The WDT has two start modes:

- Auto start mode, in which counting automatically starts after a release from the reset state
- Register start mode, in which counting starts with a refresh by writing to the register.

In auto start mode, counting automatically starts after a release from the reset state according to the settings in the Option Function Select register OFS0 or OFS3/OFS3_SEC in the flash.

In register start mode, counting starts with a refresh by writing to the WDTRR register after the respective registers are set after a release from the reset state.

Select auto start mode or register start mode by setting the Select bit (OFS0.WDT0STRT or OFS3.WDT1STRT).

When the auto start mode is selected, the settings in the WDT Control Register (WDTCR), WDT Reset Control Register (WDTRCR), and WDT Count Stop Control Register (WDTCSTPR) are disabled while the settings in the OFSm register are enabled (m = 0, 3).

When the register start mode is selected, the setting for the OFS0 or OFS3/OFS3_SEC register is disabled while the settings for the WDT Control Register (WDTCR), WDT Reset Control Register (WDTRCR), and WDT Count Stop Control Register (WDTCSTPR) are enabled.

27.3.1.1 Register Start Mode

When the WDT Start Mode Select bit (OFS0.WDT0STRT or OFS3.WDT1STRT) is 1, register start mode is selected and the WDT Control Register (WDTCR), WDT Reset Control Register (WDTRCR), and WDT Count Stop Control Register (WDTCSTPR) are enabled.

After the reset state is released, set the following:

- Clock division ratio in the WDTCR register
- Window start and end positions in the WDTCR register
- Timeout period in the WDTCR register
- Reset output or interrupt request output in the WDTRCR register
- Counter stop control during transitions to Sleep mode or Deep Sleep mode in the WDTCSTPR register

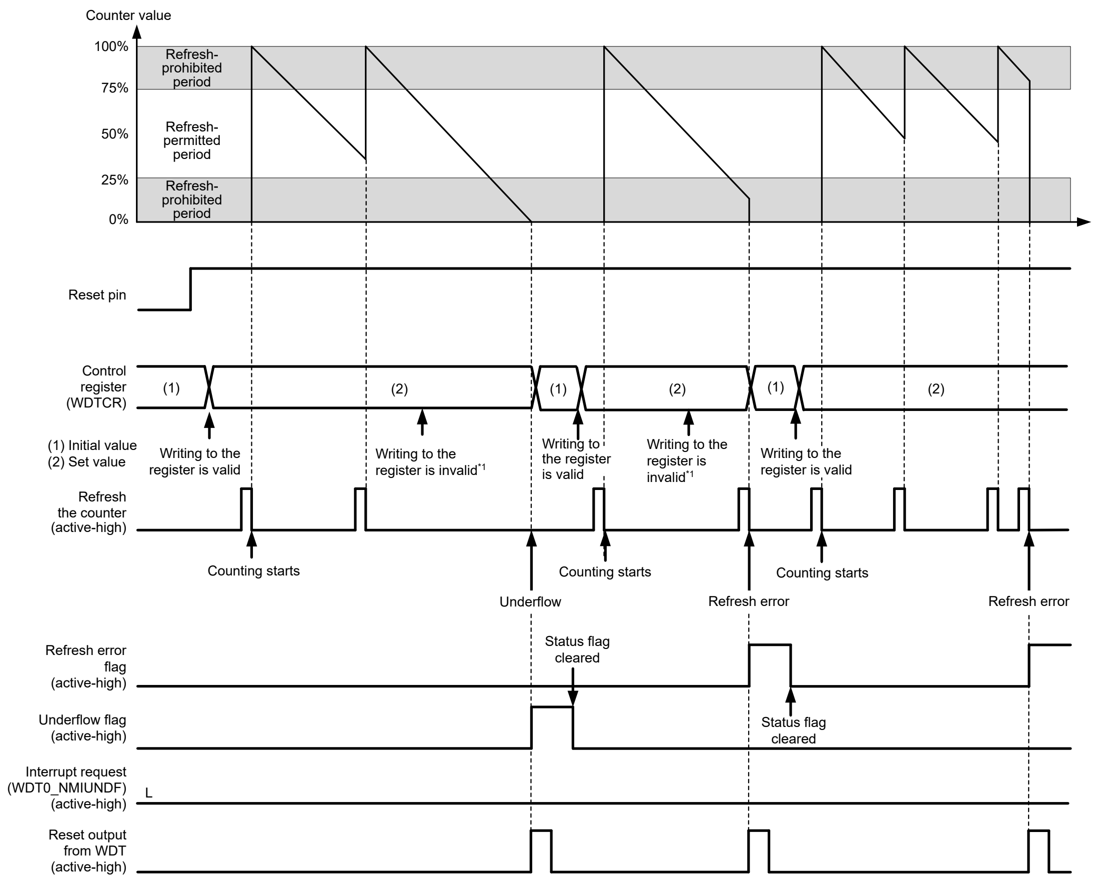
The WDT refresh register (WDTRR) refreshes the down counter. As a result, it starts counting down at the value set by the timeout period selection bit (WDTCR.TOPS[1:0]).

Thereafter, as long as the counter is refreshed in the refresh-permitted period, the value in the counter is reset each time the counter is refreshed and counting down continues. The WDT does not output the reset signal or non-maskable interrupt request/interrupt request as long as counting continues. However, if the down-counter underflows because the down-counter cannot be refreshed due to a program runaway, or if a refresh error occurs because the counter was refreshed outside the refresh-permitted period, the WDT outputs the reset signal or a non-maskable interrupt request/interrupt request (WDTn_NMIUNDF) (n = 0, 1). Reset output or interrupt request output can be selected in the WDT Reset Interrupt Request

Select bit (WDTRCR.RSTIRQS). The interrupt enabled for operating the NMI can be selected in the WDT Underflow/Refresh Error Interrupt Enable bit (NMIER.WDTEN).

Figure 27.3 shows an example of operation under the following conditions:

- Register start mode (OFS0.WDT0STRT or OFS3.WDT1STRT = 1)
- WDT reset interrupt request selection (WDTRCR.RSTIRQS = 1)
- The window start position is 75% (WDTCR.RPSS[1:0] = 10b)
- The window end position is 25% (WDTCR.RPES[1:0] = 10b)



Note: Refreshing the counter requires up to 4 count signal cycles from refresh request to execution.
Note 1. See section 27.3.2. Controlling Writes to the WDTCR, WDTRCR, and WDTCSTPR Registers.
Reset pin = RES

Figure 27.3 Operation example in register start mode

27.3.1.2 Auto Start Mode

When the WDT Start Mode Select bit (OFS0.WDT0STRT or OFS3.WDT1STRT = 1) in the Option Function Select Register m (OFSm) is 0, auto start mode is selected, the WDT Control Register (WDTCR), WDT Reset Control Register (WDTRCR), and WDT Count Stop Control Register (WDTCSTPR) are disabled, and the settings in the OFS0 or OFS3 register are enabled.

Within the reset state, the setting values for the following in the Option Function Select Register m (OFS0 or OFS3) are set in the WDT registers:

- Clock division ratio

- Window start and end positions
- Timeout period
- Reset output or interrupt request
- Counter stop control during transition to Sleep mode or Deep Sleep mode

When the reset state is released, the down-counter automatically starts counting down from the value set in the WDT Timeout Period Select bits (OFS0.WDT0STRT[1:0] or OFS3.WDT1STRT[1:0]).

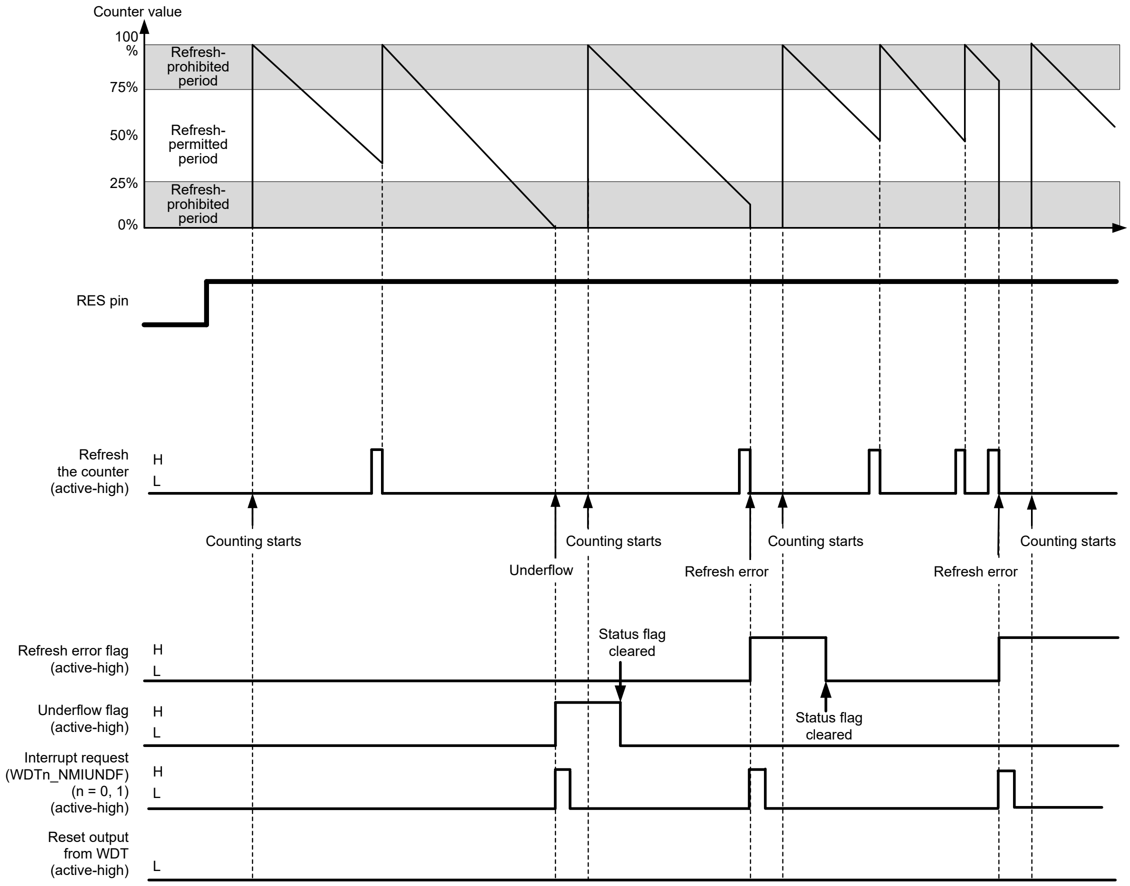
Thereafter, as long as the counter is refreshed in the refresh-permitted period, the value in the counter is reset each time the counter is refreshed and counting down continues. The WDT does not output the reset signal or non-maskable interrupt request/interrupt request (WDT_NMIUNDF) as long as the counting continues. However, if the down-counter underflows because refreshing of the down-counter is not possible due to a runaway program or if a refresh error occurs due to refreshing outside the refresh-permitted period, the WDT outputs the reset signal or non-maskable interrupt request/interrupt request (WDTn_NMIUNDF) (n = 0, 1).

After the reset signal or non-maskable interrupt request/interrupt request is generated, the counter reloads the timeout period after counting for 1 cycle. The value of the timeout period is set in the down-counter and counting restarts.

Reset output or interrupt request output can be selected by setting the WDT Reset Interrupt Request Select bit (OFS0.WDT0STRT or OFS3.WDT1STRT = 1). Non-maskable interrupt request or interrupt request can be selected in the WDT Underflow/ Refresh Error Interrupt Enable bit (NMIER.WDTEN).

Figure 27.4 shows an example of operation (non-maskable interrupt) under the following conditions:

- Auto start mode (OFSm.WDTnSTRT = 0) ((m,n) = (0,0), (3,1))
- Non-maskable interrupt request output is enabled (OFSm.WDTnRSTIRQS = 0) ((m,n) = (0,0), (3,1))
- The window start position is 75% (OFSm.WDTnRPSS[1:0] = 10b) ((m,n) = (0,0), (3,1))
- The window end position is 25% (OFSm.WDTnRPES[1:0] = 10b) ((m,n) = (0,0), (3,1))



Note: Refreshing the counter requires up to 4 count signal cycles from refresh request to execution.
Note: Reset pin = RES

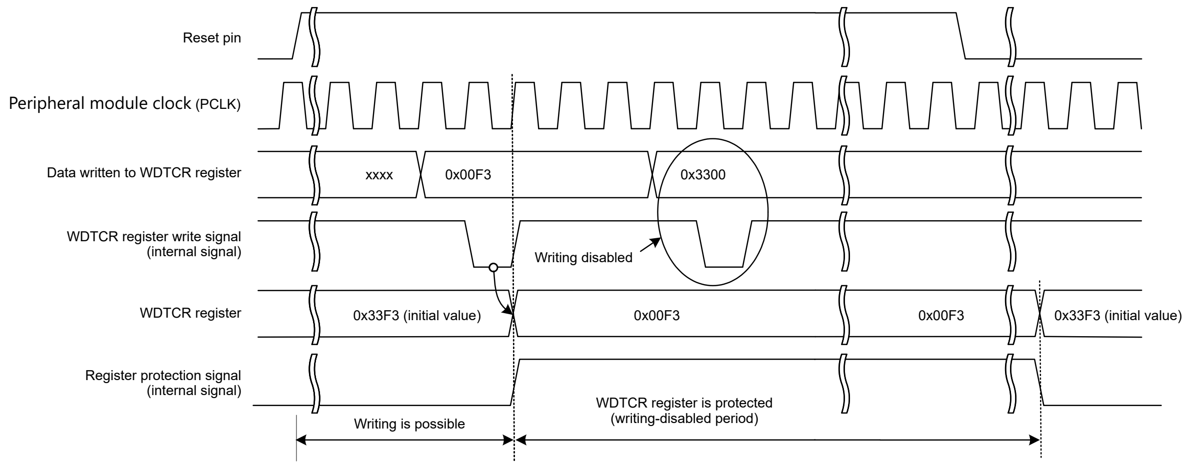
Figure 27.4 Operation example in auto start mode

27.3.2 Controlling Writes to the WDTCR, WDTRCR, and WDTCSTPR Registers

Writing to the WDT Control Register (WDTCR), WDT Reset Control Register (WDTRCR), or WDT Count Stop Control Register (WDTCSTPR) is possible once each between the release from the reset state and the first refresh operation.

After a refresh (counting starts) or a write to WDTCR, WDTRCR or WDTCSTPR register, the protection signal in the WDT becomes 1 to protect WDTCR, WDTRCR and WDTCSTPR register against subsequent write attempts. This protection is released by the reset source of the WDT. With other reset sources, the protection is not released.

Figure 27.5 shows control waveforms produced in response to writing to the WDTCR.



Note: PCLK = PCLKB
Reset pin = RES

Figure 27.5 Control waveforms produced in response to writes to the WDTCR register

27.3.3 Refresh Operation

To refresh the down counter and start the counting operation, write to the WDT Refresh Register (WDTRR) in the order of values from 0x00 to 0xFF. If a value other than 0xFF is written after 0x00, the down-counter is not refreshed. If an invalid value is written, refreshing is performed normally by writing to the WDTRR register in the order of values from 0x00 to 0xFF.

Correct refreshing is also performed when a register other than WDTRR is accessed or WDTRR is read between writing 0x00 and writing 0xFF to WDTRR. Writes to refresh the counter must be made within the refresh-permitted period, and this is determined by the 0xFF write. For this reason, correct refreshing is performed even when 0x00 is written outside the refresh-permitted period.

[Example write sequences that are valid for refreshing the counter]

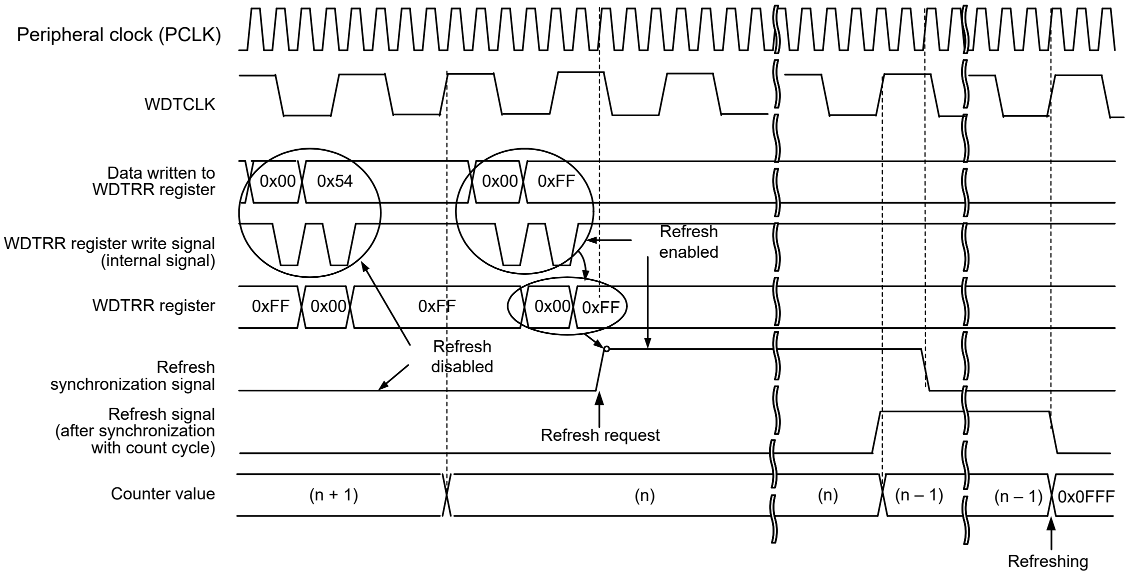
- 0x00 → 0xFF
- 0x00 ((n–1)th time) → 0x00 (nth time) → 0xFF
- 0x00 → access to another register or read from WDTRR → 0xFF

[Example write sequences that are invalid for refreshing the counter]

- 0x23 (a value other than 0x00) → 0xFF
- 0x00 → 0x54 (a value other than 0xFF)
- 0x00 → 0xAA (0x00 and a value other than 0xFF) → 0xFF

After 0xFF is written to the WDT Refresh Register (WDTRR), refreshing the down-counter requires up to 4 cycles of the signal for counting. To meet this requirement, complete writing 0xFF to WDTRR 4 count cycles before the down-counter underflows.

Figure 27.6 shows the WDT refresh-operation waveforms when the clock division ratio is PCLKB/64.



Note: PCLK = PCLKB

Figure 27.6 WDT refresh operation waveforms when WDTCR.CKS[3:0] = 0x4 and WDTCR.TOPS[1:0] = 01b

Note: When setting the refresh time, consider the oscillation accuracy of the clock sources of the PCLKB and WDTCLK. Set values which ensure that refreshing is possible even when the frequency varies in the range of error of the oscillation accuracy.

27.3.4 Status Flags

The refresh error (WDTSR.REFEF) and underflow (WDTSR.UNDFF) flags retain the source of the interrupt request from the WDT. After a release from the interrupt request generation, read the WDTSR.REFEF and WDTSR.UNDFF flags to check for the interrupt source. For each flag, writing 0 clears the bit. Writing 1 has no effect. Leaving the status flags unchanged does not affect operation. If the flags are not cleared at the next interrupt request from the WDT, the earlier interrupt source is cleared and the new interrupt source is written. For the time period between when 0 is written in each flag and when its value is reflected, see section 27.2.3. WDTSR : WDT Status Register.

27.3.5 Reset Output

When the Reset Interrupt Select bit (WDTRCR.RSTIRQS) is set to 1 in register start mode, or when the WDT Reset Interrupt Request Select bit (OFS0.WDT0RSTIRQS or OFS3.WDT1RSTIRQS) in the Option Function Select Register m (OFS0 or OFS3) is set to 1 in auto start mode, a reset signal is output for 1 cycle count when an underflow in the down-counter or a refresh error occurs.

In register start mode, the down-counter is initialized (all bits set to 0) and stopped in that state after output of a reset signal. After the reset state is released and the program is restarted, the counter is set up again and counting down starts again with a refresh. In auto start mode, counting down starts automatically after the reset state is released.

27.3.6 Interrupt Sources

When the Reset Interrupt Select bit (WDTRCR.RSTIRQS) is set to 0 in register start mode or when the WDT Reset Interrupt Request Select bit (OFS0.WDT0RSTIRQS or OFS3.WDT1RSTIRQS) in the Option Function Select Register m (OFS0 or OFS3) is set to 0 in auto start mode, an interrupt (WDTn_NMIUNDF) signal is generated when an underflow in the counter or a refresh error occurs. This interrupt can be used as a non-maskable interrupt or an interrupt. For details, see section 14, Interrupt Controller Unit (ICU).

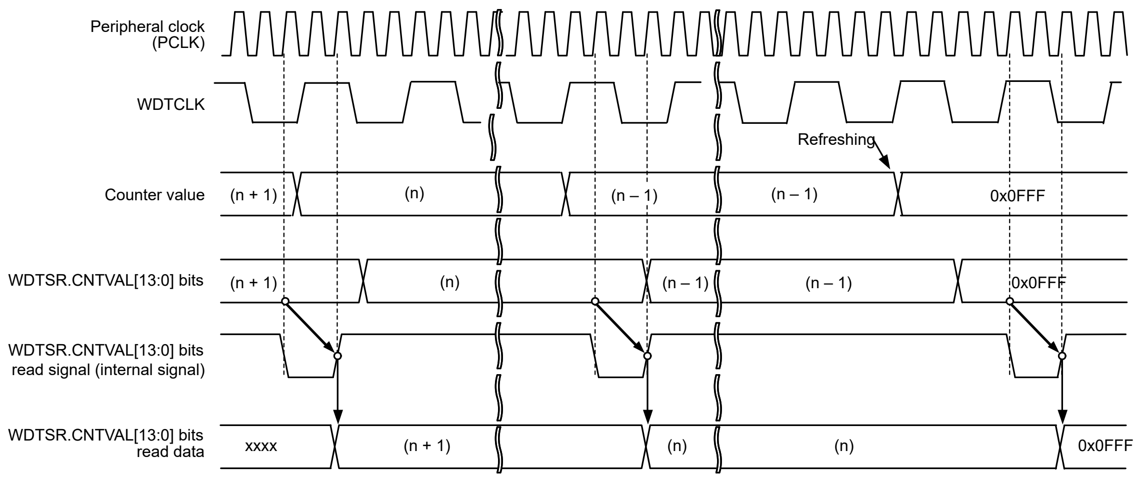
Table 27.4 WDT interrupt source

Name	Interrupt source	Interrupt to CPU	Start DMAC or DTC
WDTn_NMIUNDF (n = 0, 1)	• Down-counter underflow • Refresh error	Possible	Not possible

27.3.7 Reading the Down-Counter Value

The WDT stores the counter value in the down-counter value bits (WDTSR.CNTVAL[13:0]) of the WDT Status Register. Check these bits to obtain the counter value. The read value of the down-counter might differ from the actual count by one.

Figure 27.7 shows the processing for reading the WDT down-counter value when the clock division ratio is PCLKB/64.



Note: PCLK = PCLKB

Figure 27.7 Processing for reading WDT down-counter value when WDTCR.CKS[3:0] = 0x4 and WDTCR.TOPS[1:0] = 01b

27.3.8 Association Between Option Function Select Register m (OFSm) and WDT Registers (m = 0, 3)

Table 27.5 lists the association between the Option Function Select Register m (OFSm) used in auto start mode, and the registers used in register start mode. For details on the Option Function Select Register m (OFSm), see section 27.2.6. Option Function Select Register.

Table 27.5 Association between Option Function Select Register m (OFSm) and the WDT registers

Control target	Function	OFSm register (enabled in auto start mode) OFSm.WDTnSTRT = 0	WDT registers (enabled in register start mode) OFSm.WDTnSTRT = 1
Down-counter	Timeout period selection	OFSm.WDTnTOPS[1:0]	WDTCR.TOPS[1:0]
	Clock division ratio selection	OFSm.WDTnCKS[3:0]	WDTCR.CKS[3:0]
	Window start position selection	OFSm.WDTnRPSS[1:0]	WDTCR.RPSS[1:0]
	Window end position selection	OFSm.WDTnRPES[1:0]	WDTCR.RPES[1:0]
Reset output or interrupt request output	Select a reset interrupt request	OFSm.WDTnRSTIRQS	WDTRCR.RSTIRQS
Count stop	Sleep mode or Deep Sleep mode count stop control	OFSm.WDTnSTPCTL	WDTCSTPR.SLCSTP

27.4 Output to the Event Link Controller (ELC)

The WDT is capable of a link operation for the previously specified module when interrupt request signal is used as an event signal by the ELC. The event signal is output by the counter underflow and refresh error. An event signal is output regardless of the setting of the Reset Interrupt Request Select bit (WDTRCR.RSTIRQS) in register start mode or auto start mode. An event signal can also be output when the next interrupt source is generated while the Refresh Error flag (WDTSR.REFEF) or Underflow flag (WDTSR.UNDFF) is 1. For details, see section 19, Event Link Controller (ELC).

27.5 Usage Notes

27.5.1 ICU Event Link Setting Register n (IELSRn) Setting

Setting 0x53 to ICU Event Link Setting Register n (ICU.IELSRn) is prohibited when enabling the WDT reset assertion (OFS0.WDT0RSTIRQS or OFS3.WDT1RSTIRQS = 0 or WDTRCR.RSTIRQS = 0) or when enabling event link operation (ELSRn.ELS[8:0] = 0x93 or 0x94).